Use Cases

Version 1.0

HealthMaxxin

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Use Cases

USE CASE UC1: Create Treatment / Hospital / Department	
Goal in Context	To allow the System Admin to create basic entities for hospital structures.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	A new Treatment / Hospital / Department is created.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Admin asks the system to create a new Treatment / Hospital / Department.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR0.1	Treatment creation
FR1.1	Hospital creation
FR1.4	Department creation
NFR4	Scalability and capacity

DESCRIPTION

Step	Action
1	The System Admin asks the system to create a new Treatment / Hospital / Department.
2	The System Admin fills in the required fields.
3	The System Admin saves.
4	The system checks if the "name" field is not duplicate.
5	The system creates the new Treatment / Hospital / Department in the database.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the creation process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies a duplicated name. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC2: Create Insurance Policy	
Goal in Context	To enable the System Administrator to create new insurance policies shared across all facilities in the healthcare network.
Scope & Level	Scope: Global Insurance Policy handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	A new Insurance Policy is created.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Admin asks the system to create a new Insurance Policy.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR9.1	Create insurance policy
NFR4	Scalability and capacity

DESCRIPTION

Step	Action
1	The System Admin asks the system to create a new Insurance Policy.
2	The System Admin fills in the required fields.
3	The System Admin saves.
4	The system checks if the discount percentage is a valid amount.
5	The system creates the new Insurance Policy in the database.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the creation process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies an invalid amount for the discount percentage. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC3: Create Prescription / Exemption	
Goal in Context	To enable a doctor to create a new Prescription / Exemption for a specific patient.
Scope & Level	Scope: Prescriptions / Exemptions. Level: Primary.
Preconditions	The doctor is logged into the system.
Success End Condition	A new Prescription / Exemption is created and linked to the patient's profile.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: Doctor. Secondary: Patient, Database, Mail Service.
Trigger	The doctor asks the system to create a new Prescription / Exemption.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR29	Prescription creation
FR30	Exemption creation
NFR4	Scalability and capacity

DESCRIPTION

Step	Action
1	The doctor asks the system to create a new Prescription / Exemption.
2	The doctor fills in the required fields, including the selection of the patient to whom the Prescription / Exemption applies.
3	The doctor saves.
4	The system verifies that the selected patient is valid.
5	The system creates the new Prescription / Exemption in the database.
6	The system returns a success message.
7	The patient is notified via email that the Prescription / Exemption has been linked to their account.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the creation process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies an invalid value for the patient field. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC4: Upload examination	
Goal in Context	To allow a doctor to upload and register a completed medical examination, including diagnosis, documents, and notes.
Scope & Level	Scope: Clinical Records Management. Level: Primary.
Preconditions	A medical examination has been performed and the doctor is logged into the system..
Success End Condition	The examination is successfully uploaded and becomes visible in the patient's dashboard.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: Doctor. Secondary: Patient, Database, Mail Service.
Trigger	The doctor asks the system to upload an examination to the patient clinical record.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR33	Examination upload to patient clinical record
NFR4	Scalability and capacity

DESCRIPTION

Step	Action
1	The doctor asks the system to upload an examination to the patient clinical record.
2	The doctor fills in the required fields, including the selection of the patient to whom the Prescription / Exemption applies.
3	The doctor appends as much attachments as wanted.
4	The doctor saves.
5	The system perform field validations.
6	The system saves the examination record in the patient clinical record.
7	The system returns a success message.
8	The patient is notified via email that a new examination has been uploaded to their record

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the creation process. 2. Use case ends.
5a	**Validity check fails:** 1. The system identifies an invalid value for the patient field. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.
5b	**Invalid attachment type:** 1. The system identifies an invalid attachment type. 2. The system returns an error. 3. Go to main flow step 3 to perform a correction.

USE CASE UC5: Edit Treatment / Hospital / Department	
Goal in Context	To allow the System Admin to edit basic entities for hospital structure.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	An existing Treatment / Hospital / Department is edited.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Admin asks the system to create a new Treatment / Hospital / Department.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR0.2	Treatment editing
FR1.2	Hospital editing
FR1.5	Department editing

DESCRIPTION

Step	Action
1	The System Admin asks the system to edit an existing Treatment / Hospital / Department.
2	The System Admin fills in the required fields.
3	The System Admin saves.
4	The system checks if the "name" field is not duplicate.
5	The system edits the Treatment / Hospital / Department in the database.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the editing process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies a duplicated name. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC6: Update Insurance Policy	
Goal in Context	To enable the System Administrator to edit an existing insurance policies shared across all facilities in the healthcare network.
Scope & Level	Scope: Global Insurance Policy handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	An existing Insurance Policy is edited.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System admin. Secondary: Database.
Trigger	The System Admin asks the system to edit an existing Insurance Policy.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR9.3	Update insurance policy

DESCRIPTION

Step	Action
1	The System Admin asks the system to update an existing Insurance Policy.
2	The System Admin fills in the required fields.
3	The System Admin saves.
4	The system checks if the discount percentage is a valid amount.
5	The system updates the Insurance Policy in the database.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the editing process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies an invalid amount for the discount percentage. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC7: Update User Profile	
Goal in Context	To provide a Registered user with to modify their account information.
Scope & Level	Scope: users account management. Level: Primary.
Preconditions	The user is logged in.
Success End Condition	The account information is successfully updated in the system.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: User. Secondary: Database.
Trigger	The authenticated user wants to update their account.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR13.2	Update user profile

DESCRIPTION

Step	Action
1	The authenticated user wants to update their account.
2	The user edits some fields.
3	The user saves.
4	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The user cancels:** 1. The system aborts the editing process. 2. Use case ends.

USE CASE UC8: Deactivate or close Treatment / Hospital / Department	
Goal in Context	To allow the System Admin to deactivate or close basic entities for hospital structure.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	An existing Treatment / Hospital / Department is deactivated or closed.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Admin asks the system to deactivate or close a new Treatment / Hospital / Department.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR0.3	Treatment deactivation
FR1.3	Hospital closure
FR1.6	Department closure

DESCRIPTION

Step	Action
1	The System Admin asks the system to deactivate or close an existing Treatment / Hospital / Department.
2	The System asks for confirmation.
3	The System Admin confirms.
4	The system marks the Treatment / Hospital / Department as deactivated or closed.
5	The system records date and time of deactivation / closure.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the deactivation / closing process. 2. Use case ends.

USE CASE UC9: Deactivate Insurance Policy	
Goal in Context	To enable the System Administrator to deactivate an existing insurance policies shared across all facilities in the healthcare network.
Scope & Level	Scope: Global Insurance Policy handling. Level: Primary.
Preconditions	The user is logged in as a system admin.
Success End Condition	An existing Insurance Policy is deactivated.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: System admin. Secondary: Database.
Trigger	The System Admin asks the system to deactivate an existing Insurance Policy.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR9.4	Delete insurance policy

DESCRIPTION

Step	Action
1	The System Admin asks the system to deactivate an existing Insurance Policy.
2	The system asks for confirmation.
3	The System Admin confirms.
4	The system marks the Insurance Policy as deactivated.
5	The system records date and time of deactivation.
6	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the editing process. 2. Use case ends.

USE CASE UC10: Cancel appointment (Patient)	
Goal in Context	To enable patient to cancel a booked medical appointment
Scope & Level	Scope: Appointment management. Level: Primary.
Preconditions	The patient is logged in.
Success End Condition	An existing medical appointment is deleted.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: Patient. Secondary: Database, Payment Service, Mail Service.
Trigger	Patient asks the system to cancel a booked appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR16.2	Cancel appointment (patient)

DESCRIPTION

Step	Action
1	Patient asks the system to cancel a booked appointment.
2	The system asks for confirmation.
3	The patient confirms.
4	The system performs validity check.
5	The system cancels the appointment.
6	The patient is refounded.
7	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The patient cancels:** 1. The system aborts the editing process. 2. Use case ends.
4a	**Late canceling:** 1. The system identifies that the appointment is in less than 48 hours. 2. Cancellation is aborted. 3. The system returns an error. 4. Use case ends.
6a	**Appointment not payed:** 1. The system checks if the appointment was paid ahead. 2. If not go to main flow step 7.

USE CASE UC11: Cancel appointment (Hospital staff)	
Goal in Context	To enable doctor / head of department to cancel a booked medical appointment
Scope & Level	Scope: Appointment management. Level: Primary.
Preconditions	The doctor / head of department is logged in.
Success End Condition	An existing medical appointment is deleted.
Failed End Condition	The system remains unchanged. The system returns an error.
Primary, Secondary Actors	Primary: Doctor / Head of department. Secondary: Database, Payment Service, Mail Service.
Trigger	The Doctor / Head of department asks the system to cancel a booked appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR35.2	Cancel appointment (Doctor)
FR36.2	Cancel appointment (Head of department)

DESCRIPTION

Step	Action
1	The doctor / head of department asks the system to cancel a booked appointment.
2	The doctor / head of department provides a cancellation reason.
3	The doctor / head of department confirms.
4	The system performs validity check.
5	The system cancels the appointment.
6	The patient is notified via email about the appointment cancellation with the reason.
7	The patient is refounded.
8	The system returns a success message.

EXTENSIONS

Step	Branching Action
3a	**The doctor / head of department cancels:** 1. The system aborts the editing process. 2. Use case ends.
4a	**Late canceling:** 1. The system identifies that the appointment is in less than 48 hours. 2. Cancellation is aborted. 3. The system returns an error. 4. Use case ends.
7a	**Appointment not payed:** 1. The system checks if the appointment was paid ahead. 2. If not go to main flow step 7.

USE CASE UC12: Register Doctor	
Goal in Context	A System Admin wants to add a new doctor to the hospital's digital infrastructure.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a System Admin. The relevant hospital structure already exists.
Success End Condition	The doctor is successfully saved in the database, and a set of one-time credentials is provided to the System Admin.
Failed End Condition	The registration is aborted due to invalid data or duplication; no credentials are created.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Admin asks the system to create a new doctor account.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR1.7	Doctor registration

DESCRIPTION

Step	Action
1	The System Admin asks the system to create a new doctor account.
2	The System Admin fills in the required fields, including doctor's personal data.
3	The System Admin saves.
4	The system checks if the "name" field is not duplicate.
5	The system creates the new doctor acount in the database.
6	The system returns the new account credentials, which are diplayed just once.

EXTENSIONS

Step	Branching Action
3a	**The System Admin cancels:** 1. The system aborts the creation process. 2. Use case ends.
4a	**Validity check fails:** 1. The system identifies duplicated personal data for the doctor. 2. The system returns an error. 3. Go to main flow step 2 to perform a correction.

USE CASE UC13: Assign Doctor	
Goal in Context	To enable the System Administrator to assign a Registered Doctor to operate within a specific hospital structure.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged as a System Admin. The specific hospital has been created. The doctor is registered in the system.
Success End Condition	The selected doctor is successfully assigned to the specified hospital.
Failed End Condition	Doctor assignment fails. The previous assignment remains unchanged.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	System Admin wants to assign a doctor to a specific hospital structure.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR1.8	Doctor assignment

DESCRIPTION

Step	Action
1	System Admin wants to assign a doctor to a specific hospital structure.
2	System admin picks the selected doctor from the set of registered Doctors.
3	System admin saves.
4	The system asks for confirmation.
5	System admin confirms.
6	The System performs validity checks.
7	The doctor is assigned to the hospital and the system status is updated.

EXTENSIONS

Step	Branching Action
5a	**Cancel operation:** 1. System admin cancels. 2. No changes are made to the system. 3. Use case ends.
6a	**Doctor already assigned:** 1. The System finds the specified doctor already assigned. 2. The System asks for permission to reassign already assigned doctor. 3. The System Administrator confirms. (If not, the use case ends) 4. Go to main flow step 6.

USE CASE UC14: Appoint Head of Hospital	
Goal in Context	To enable the System Administrator to appoint a Registered Doctor to the strategic role of Head of Hospital for a specific facility.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a System Administrator. The specific hospital has been created. The doctor is already registered in the system.
Success End Condition	The selected doctor is successfully appointed as Head of Hospital.
Failed End Condition	The appointment fails; the system state remains unchanged and the previous head (if any) remains in place.
Primary, Secondary Actors	Primary: System Administrator. Secondary: Database.
Trigger	The System Administrator wants to appoint a doctor as Head of Hospital.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR5	Strategic role appointment (Head of Hospital)

DESCRIPTION

Step	Action
1	The System Administrator wants to appoint a doctor as Head of Hospital.
2	The Admin selects the desired doctor from the list of Registered Doctors.
3	The Admin saves.
4	The System prompts the Admin with a confirmation message.
5	The Admin confirms.
6	The System performs validity checks.
7	The doctor is appointed as Head of Hospital and the System status is updated.

EXTENSIONS

Step	Branching Action
2a	**Already existing Head of Hospital:** 1. The system prompts that there is another Head of Hospital for that structure. 2. The system ask if the admin wants to change the Head of Hospital. 3. Admin confirms. 4. Resume at Main Flow Step 6.
5a	**Operation canceled:** 1. Admin cancels the operation. 2. No changes are made to the database. 3. Use case ends.
6a	**Doctor not found:** 1. System cannot find the selected doctor. 2. Operation is aborted. 3. System returns an error. 4. Use case ends.

USE CASE UC15: Appoint Head of Department	
Goal in Context	To enable the Head of Hospital to appoint a Registered Doctor to the tactical role of Head of Department within their facility.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged as a Head of Hospital. The specific department has been created. The doctor is registered in the system.
Success End Condition	The selected doctor is successfully appointed as head of department.
Failed End Condition	Head of department appointment fails. The system remains unchanged.
Primary, Secondary Actors	Primary: Head of Hospital. Secondary: Database.
Trigger	The Head of Hospital wants to appoint a doctor as Head of Department.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR3	Tactical role appointment (Head of Department)

DESCRIPTION

Step	Action
1	The Head of Hospital wants to appoint a doctor as Head of Department.
2	The Head of Hospital picks the selected doctor from the set of Registered Doctors in their facility.
3	The Head of Hospital saves.
4	The Head of Hospital is prompted with a confirmation message.
5	The Head of Hospital confirms.
6	The System performs validity checks.
7	The doctor is appointed as head of department and the System status is updated.

EXTENSIONS

Step	Branching Action
2a	**Already existing Head of Department:** 1. The system prompts that there is another Head of Department for that department. 2. The system ask if the Head of Hospital wants to change the Head of Department. 3. Head of Hospital confirms. 4. Resume at Main Flow Step 6.
5a	**Operation canceled:** 1. Head of Hospital cancels the operation. 2. No changes are made to the database. 3. Use case ends.
6a	**Doctor not found:** 1. System cannot find the selected doctor. 2. Operation is aborted. 3. System returns an error. 4. Use case ends.

USE CASE UC16: Allocate facility's doctors	
Goal in Context	To enable the Head of Hospital to allocate operational resources to a specific department within their facility.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a Head of Hospital. The specific departments have been created. The specific doctors are registered in the system.
Success End Condition	The selected doctors are successfully allocated to the selected department.
Failed End Condition	The doctors' allocation fails; the previous allocation remains unchanged.
Primary, Secondary Actors	Primary: Head of Hospital. Secondary: Database.
Trigger	The Head of Hospital wants to allocate doctors for a specific facility.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR4	Facility Doctors allocation

DESCRIPTION

Step	Action
1	The Head of Hospital wants to allocate doctors for a specific facility.
2	The Head of Hospital picks the selected doctors from the set of Registered Doctors within the system.
3	The Head of Hospital saves the selection.
4	The System performs validity checks.
5	The doctors are allocated to the department and the System status is updated.

EXTENSIONS

Step	Branching Action
4a	**Doctors already allocated:** 1. The System identifies that the specified doctors are already allocated to another department. 2. The System asks for permission to reassign the already allocated doctors. 3. The Head of Hospital confirms (if not, the use case ends). 4. Resume at Main Flow Step 6.

USE CASE UC17: Configure Department's services	
Goal in Context	To enable the Head of Department to configure and update the list of medical treatments offered by their specific department.
Scope & Level	Scope: Managing the global hospital catalog. Level: Primary.
Preconditions	The hospital and department have been created, and the global catalog has been populated.
Success End Condition	The offered treatments for the specific department are successfully updated in the system.
Failed End Condition	The changes are not saved; the department's treatment offerings remain in their previous state.
Primary, Secondary Actors	Primary: Head of Department. Secondary: Database, Global Treatment Catalog.
Trigger	The Head of Department wants to configure the list of medical treatments offered for their department.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR5	Departmental service configuration

DESCRIPTION

Step	Action
1	The Head of Department wants to configure the list of medical treatments offered for their department.
2	The System displays all available treatments retrieved from the global catalog.
3	The Head of Department disables or enables a specific treatment.
4	The Head of Department clicks saves.
5	The System performs validity checks.
6	The System updates the department records and confirms the changes to the user.

EXTENSIONS

Step	Branching Action
5a	**Data validation fails:** 1. The System cannot find selected treatment in the database. 2. Operation is aborted. 3. The system prompts an error message. 4. Use case ends.

USE CASE UC18: View Insurances' Debts	
Goal in Context	To enable the Head of Hospital to view and monitor the outstanding financial debt owed by insurance companies to their facility.
Scope & Level	Scope: Managing insurance. Level: Primary.
Preconditions	The user is logged in with Head of Hospital privileges.
Success End Condition	The Head of Hospital successfully views the current debt breakdown.
Failed End Condition	The system fails to retrieve financial records; the Head of Hospital is notified of the data unavailability.
Primary, Secondary Actors	Primary: Head of Hospital. Secondary: Database.
Trigger	The Head of Hospital wants to monitor the outstanding debt owned by insurance companies to their facility.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR6.1	Insurance debt dashboard

DESCRIPTION

Step	Action
1	The Head of Hospital wants to monitor the outstanding debt owned by insurance companies to their facility.
2	The System displays a list of debt amounts categorized by each individual insurance company.
3	The Head of Hospital reviews the high-level financial summary.

EXTENSIONS

Step	Branching Action
1a	**Data not found:** 1. The system fails to retrieve insurance data. 2. The Head of Hospital is prompted with an error message. 3. Use case ends.
2a	**View historical records:** 1. The Head of Hospital wants to monitor historical records. 2. The System displays a detailed historical debt record.

USE CASE UC19: Insurance Debt History	
Goal in Context	To enable the Head of Hospital to view the detailed information of a specific insurance company's debt history regarding their facility.
Scope & Level	Scope: Managing insurance. Level: Primary.
Preconditions	The user is logged in as Head of Hospital, the insurance company has some insurance policy registered in the system.
Success End Condition	The system successfully retrieves and displays the debt history for the specific insurance company.
Failed End Condition	The system is unable to display information and returns an error.
Primary, Secondary Actors	Primary: Head of Hospital. Secondary: Database.
Trigger	The Head of Hospital wants to see the debt history for a specific insurance company.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR6.2	Insurance Debt History

DESCRIPTION

Step	Action
1	The Head of Hospital selects a specific insurance company to view its debt history.
2	The system validates the user's access rights for this specific action.
3	The system queries the Database to retrieve the current outstanding debt.
4	The system queries the Database to retrieve previous debt settlements along with their associated dates and amounts.
5	The system displays the comprehensive debt history to the Head of Hospital.

EXTENSIONS

Step	Branching Action
2a	**Access validation fails:** 1. The system identifies that the user lacks the specific rights to view this detailed record. 2. The system prompts an error. 3. The view process is aborted. Use case ends.
3a	**Entity not found:** 1. The system is unable to locate the insurance company in the database. 2. The system prompts an error message explaining that the entity is no longer available. 3. Use case ends.
4a	**No previous debt history:** 1. The system finds no past settlements for the selected insurance company. 2. The system displays an error. 3. Use case ends.

USE CASE UC20: Register insurance payment	
Goal in Context	To enable the Head of Hospital to manually settle and clear the outstanding financial balance owed by a specific insurance company to their facility.
Scope & Level	Scope: Managing insurance. Level: Primary.
Preconditions	The user is logged in as the Head of Hospital. The hospital exists, an outstanding debt is present, and the insurance company has paid out the debt.
Success End Condition	The outstanding debt is marked as settled, and the transaction is recorded in the system.
Failed End Condition	The settlement process is cancelled or fails; the debt remains unchanged.
Primary, Secondary Actors	Primary: Head of Hospital. Secondary: Financial Database.
Trigger	The Head of Hospital wants to settle the outstanding debt for a specific insurance company.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR7	Manual settlement of insurance debt

DESCRIPTION

Step	Action
1	The Head of Hospital wants to settle the outstanding debt for a specific insurance company.
2	The Head of Hospital fills in the payment data.
3	The Head of Hospital saves.
4	The System validates the input fields.
5	The System displays a confirmation alert showing the exact amount being settled.
6	The Head of Hospital confirms.
7	The System updates the database to reflect the settled status and adjusts the total debt.
8	The System provides a success notification.

EXTENSIONS

Step	Branching Action
4a	**Validity check fails:** 1. The system identifies an invalid field for the payment details. 2. The system displays an error message . 3. Resume at Main Flow Step 2.
6a	**Head of Hospital cancels:** 1. No changes are made to the database. 2. The settlement process is immediately terminated. 3. Use case ends.

USE CASE UC21: Departmental operational management	
Goal in Context	To enable the Head of Department to configure the operational capacity of the department.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a Head of Department.
Success End Condition	The operational capacity is successfully updated.
Failed End Condition	The operational capacity remains unchanged.
Primary, Secondary Actors	Primary: Head of Department. Secondary: Database
Trigger	The Head of Department wants to configure operational capacity.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR8.1	Department capacity configuration

DESCRIPTION

Step	Action
1	The Head of Department wants to configure operational capacity (number and type of rooms).
2	The Head of Department updates the desired fields and saves.
3	The system validates input fields.
4	The System updates the database to reflect the new operational capacity.
5	The System provides a success notification.

EXTENSIONS

Step	Branching Action
3a	**Validity check fails:** 1. The system identifies an invalid field for the operational capacity. 2. The system displays an error message highlighting the invalid field. 3. Resume at Main Flow Step 2.

USE CASE UC22: Generate Insurance Policy code	
Goal in Context	To provide the System Administrator the ability to generate the insurance policy code.
Scope & Level	Scope: Insurance policies management. Level: Primary.
Preconditions	The insurance policy has been created and the Patient is registered in the system.
Success End Condition	The unique code is successfully created.
Failed End Condition	The unique code is not created.
Primary, Secondary Actors	Primary: System Admin. Secondary: Database.
Trigger	The System Administrator wants to generate a univoque insurance code for a specific policy.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR10	Simulated Insurance code generation

DESCRIPTION

Step	Action
1	The System Administrator wants to generate a univoque insurance code for a specific policy.
2	The System generate a code.

EXTENSIONS

Step	Branching Action
2a	**System error during generation:** 1. The System encounters a database or internal error and cannot generate the code. 2. The System displays an error message prompting to try again. 3. Use case ends.

USE CASE UC23: User registration and validation	
Goal in Context	To allow an Unregistered Patient to create and validate an account within the system
Scope & Level	Scope: Users account management. Level: Primary Task.
Preconditions	The patient is not authenticated and does not have an active account.
Success End Condition	A new account is registered in the system and successfully validated.
Failed End Condition	No new account is created, or the account remains unverified; the user is prompted with an error message.
Primary, Secondary Actors	Primary: Patient. Secondary: Database, Mail Service, Identity Provider (SPID/CIE).
Trigger	The Unregistered Patient wants to register in the system.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR11.1	User registration
FR11.2	User validation
NFR0	Security and privacy
NFR6	SPID/CIE-based registration and authentication
NFR7	Login validity period

DESCRIPTION

Step	Action
1	The System asks the Unregistered Patient to insert their email and personal information.
2	The patient fills in the required information.
3	The Unregistered Patient submits.
4	The System validates the input fields.
5	The patient is prompted to create a password.
6	The patient submits a password.
7	The System checks password security (complexity and strength).
8	The System creates a new unverified account in the database.
9	The System sends a verification token to the provided email address via the Mail Service.
10	The User accesses their mailbox and clicks on the validation link.
11	The System retrieves the validation token sent with the request.
12	The System checks for token validity and matching records in the database.
13	The System marks the account as verified and active.
14	The System automatically logs the user in and redirects them to the home page.

EXTENSIONS

Step	Branching Action
4a	**Duplicated fiscal code:** 1. The System identifies a duplicated fiscal code. 2. The patient is prompted with an error message highlighting the invalid field. 3. Resume at Main Flow Step 2.
4b	**Email already taken:** 1. The System finds another account already associated with the provided email. 2. The patient is prompted with an error message suggesting to log in or reset the password. 3. Use case ends.
6a	**Password security check fails:** 1. The System identifies the provided password as unsafe or too simple. 2. The System prompts an error message with security requirements. 3. Resume at Main Flow Step 5.
11a	**Invalid or Expired validation token:** 1. The System identifies that the token from the email link is invalid or has expired. 2. The patient is prompted with an error message and asked if they want to receive a new validation email. 3. The patient either confirms (go to Main flow step 9) or declines (use case ends).

SUB-VARIATIONS

Step	Branching Action
2a	**SPID or CIE register:** 1. The User selects to register using a National Digital Identity Provider (SPID/CIE). 2. Authentication is performed by the external authority, which returns a secure token. 3. The System maps the token to the corresponding user account. 4. Use case ends.

USE CASE UC24: User Login	
Goal in Context	To allow a registered user to securely log into their account.
Scope & Level	Scope: Account management. Level: Primary Task.
Preconditions	The user has a registered and validated account in the system.
Success End Condition	The user successfully logs in and is redirected to their dashboard.
Failed End Condition	The login fails, and the user is prompted with an error message.
Primary, Secondary Actors	Primary: User. Secondary: Database, Identity Provider (SPID/CIE).
Trigger	The user wants to log into the system.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR11.3	User login
NFR0	Security and privacy
NFR6	SPID/CIE-based registration and authentication
NFR7	Login validity period

DESCRIPTION

Step	Action
1	The User inputs their registered email and password.
2	The System verifies the provided credentials against the database.
3	The user succesfully logs in.

EXTENSIONS

Step	Branching Action
2a	**Invalid Credentials:** 1. The System identifies that the provided email or password is incorrect. 2. The System displays an error message. 3. Resume at Main Flow Step 1.
3a	**Mandatory Password Change:** 1. The System detects that the User (e.g., a newly registered Doctor) is logging in for the first time using a system-generated default password. 2. The User is prompted to update their credentials through the Account recovery use case. 3. Use case ends.

SUB-VARIATIONS

Step	Branching Action
1a	**SPID or CIE login:** 1. The User selects to log in using a National Digital Identity Provider (SPID/CIE). 2. Authentication is performed by the external authority, which returns a secure token. 3. The System maps the token to the corresponding user account. 4. Resume at Main Flow Step 3.

USE CASE UC25: Automated Doctor Assignment	
Goal in Context	To automatically assign the most suitable doctor to a newly requested appointment based on specialty and continuity of care.
Scope & Level	Scope: Doctor assignment. Level: Subfunction.
Preconditions	The calling use case has provided the examination type, requested date, hospital, and available room.
Success End Condition	A suitable doctor is successfully assigned to the appointment.
Failed End Condition	No doctor is assigned; the system notifies the Head of Department of the failure.
Primary, Secondary Actors	Primary: System. Secondary: Patient, Database, Mail Service, Head of Department.
Trigger	The use case is triggered automatically when a patient is booking or rescheduling an appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR12	Automated doctor assignment
NFR1	Algorithm performance

DESCRIPTION

Step	Action
1	The use case is triggered automatically when a patient is booking or rescheduling an appointment
2	The System filters for doctors whose specialty matches the requested examination type.
3	The System filters for doctors that have already visited the patient.
4	The System picks the doctor with the earliest availability closest to the requested date, provided they are available within a 30-day window.

EXTENSIONS

Step	Branching Action
2a	**No doctor found with required specialty:** 1. The System sends a notification to the Head of Department. 2. The System returns an error to the calling use case. 3. Use case ends.
3a	**No doctor found who has previously visited the patient:** 1. The System bypasses the continuity filter from Step 2 and considers all doctors from Step 1. 2. Resume at Main Flow Step 4.
4a	**No past doctor available within 30 days:** 1. The System bypasses the continuity filter and considers all doctors from Step 1. 2. The System picks the doctor with the earliest availability closest to the requested date. 3. If still no doctor is found, proceed to Step 4b.
4b	**No available doctor found:** 1. The System sends a notification to the Head of Department reporting the error. 2. The System returns an error to the calling use case. 3. Use case ends.

USE CASE UC26: Account recovery	
Goal in Context	To provide the user with a way to recover their account access by resetting their password.
Scope & Level	Scope: users account management. Level: Primary.
Preconditions	The user has a registered account in the system.
Success End Condition	The account password is successfully changed.
Failed End Condition	The password is not changed; the account remains inaccessible via the old credentials.
Primary, Secondary Actors	Primary: User. Secondary: Database, Mail Service.
Trigger	The user wants to recover their account access.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR14	Account recovery

DESCRIPTION

Step	Action
1	The user wants to recover their account access.
2	The user enters the email and submits.
3	The System validates the email address.
4	The System sends an email containing a secure, time-limited, password reset link.
5	The System displays a confirmation message.
6	The user clicks the link in their email.
7	The user inputs a new password and confirms.
8	The System checks the password security and complexity.
9	The user's password is updated and saved securely in the System.

EXTENSIONS

Step	Branching Action
3a	**Email not in the database:** 1. The System identifies that the email does not belong to any Registered User. 2. The System fails silently (no mail is sent) to protect user privacy. 3. Use case ends.
3b	**Wrong email format:** 1. The System identifies an invalid email format. 2. An error message is displayed. 3. Resume at Main Flow Step 1.
6a	**Invalid or expired token:** 1. If the reset link is invalid or has expired, the System displays an error message. 2. Use case ends.
8a	**Password security check fails:** 1. The System identifies the provided password as unsafe or weak. 2. The System prompts an error message. 3. Resume at Main Flow Step 7.

USE CASE UC27: Appointment booking	
Goal in Context	To provide a logged-in Patient a way to book a medical visit.
Scope & Level	Scope: Appointment management. Level: Primary.
Preconditions	The patient is logged into the system.
Success End Condition	An appointment (comprising date, time, hospital, department, and doctor) is successfully booked.
Failed End Condition	No appointment is booked; the system remains unchanged.
Primary, Secondary Actors	Primary: Patient. Secondary: Database, Mail Service.
Trigger	The patient wants to book an appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR15	Appointment booking
NFR1	Algorithm performance
NFR4	Scalability and capacity
NFR5	Accessibility
NFR8	Concurrency and data integrity

DESCRIPTION

Step	Action
1	The patient selects the desired examination and, optionally, a preferred hospital and a target future date.
2	The patient submits the search request.
3	The System selects all periods in which at least one correct examination room is available.
4	**Include: FR12 Automated doctor assignment.** The System retrieves a suitable doctor, and date and time of the appointment.
5	The System presents the proposed appointment details (date, time, hospital, department, and doctor) to the patient.
6	The patient confirms.
7	The System saves the appointment in the database.
8	The System sends an email notification to the patient.

EXTENSIONS

Step	Branching Action
2a	**Examination not available in the preferred hospital:** 1. The System displays an alert stating the examination is unavailable at the selected hospital. 2. The System automatically selects the closest hospital to the patient's registered address that offers the selected examination. 3. Resume at Main Flow Step 3.
2b	**No preferred hospital:** 1. The System automatically selects the closest hospital to the patient's registered address that offers the selected examination. 2. Resume at Main Flow Step 3.
2c	**No preferred date:** 1. The System automatically defaults to the current date as the search starting point. 2. Resume at Main Flow Step 3.
3a	**No room available found within 30 days:** 1. The System displays an error message asking the patient to select another hospital or date range. 2. Resume at Main Flow Step 1.
4a	**No suitable appointment is found:** 1. The System displays an error message asking the patient to select another hospital or date range. 2. Resume at Main Flow Step 1.
6a	**The patient cancels:** 1. No action is taken and the system remains unchanged. 2. Use case ends.

USE CASE UC28: Appointment View	
Goal in Context	To provide a patient / Doctor / Head of department a way to consult booked medical visits.
Scope & Level	Scope: Appointment management. Level: Primary.
Preconditions	The user is logged into the system as a Patient / Doctor / Head of Department.
Success End Condition	The system successfully retrieves and displays the detailed information of the appointment.
Failed End Condition	The system is unable to display the details and returns an error.
Primary, Secondary Actors	Primary: Patient / Doctor / Head of Department. Secondary: Database.
Trigger	The patient / Doctor / Head of Department wants to see the details of an appointment they are entitled to see.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR16.1	View appointment (patient)
FR35.1	View appointment (doctor)
FR36.1	View appointment (head of department)

DESCRIPTION

Step	Action
1	The patient / Doctor / Head of Department wants to see the details of an appointment they are entitled to see.
2	The system checks the user's access rights for that specific appointment.
3	The system retrieves the appointment details from the database.
4	The system displays the detailed information of the appointment to the user.

EXTENSIONS

Step	Branching Action
2a	**Access validation fails:** 1. The system identifies that the user lacks the rights to view this appointment (e.g., a Patient trying to view an appointment not linked to their account). 2. The system returns an error. 3. Use case ends.
3a	**Appointment not found:** 1. The system is unable to locate the appointment in the database. 2. The system returns an error. 3. Use case ends.

USE CASE UC29: Appointment Rescheduling	
Goal in Context	To allow a logged-in patient to reschedule an existing appointment to a new available time slot, provided they give at least 48 hours of notice.
Scope & Level	Scope: Appointment management. Level: Primary.
Preconditions	The patient is logged into the system and has a booked future appointment.
Success End Condition	The appointment is successfully rescheduled to a new date, time, and assigned doctor.
Failed End Condition	The appointment remains unchanged. The system prompts an error (e.g., ¡ 48 hours notice, no slots available).
Primary, Secondary Actors	Primary: Patient. Secondary: Database, Mail Service.
Trigger	The patient wants to reschedule a specific medical appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR16.3	Reschedule Appointment

DESCRIPTION

Step	Action
1	The patient wants to reschedule a specific medical appointment.
2	The System verifies that the scheduled appointment is at least 48 hours away.
3	The System prompts the patient to select a new target future date.
4	The Patient submits the search request for the new date.
5	The System selects all periods in which at least one correct examination room is available.
6	**Include: FR12 Automated doctor assignment.** The System retrieves a suitable doctor, and date and time of the appointment based on the new request.
7	The System presents the newly proposed appointment details (date, time, and assigned doctor) to the patient.
8	The Patient accepts.
9	The System updates the appointment record in the database with the new details.
10	The System sends an email notification to the patient confirming the new schedule.

EXTENSIONS

Step	Branching Action
2a	**Less than 48 hours notice:** 1. The System identifies that the appointment is less than 48 hours away. 2. The System blocks the action and prompts an error message stating that rescheduling is no longer allowed. 3. Use case ends.
3a	**No preferred date selected:** 1. The System automatically defaults to the current date as the search starting point. 2. Resume at Main Flow Step 5.
5a	**No room available found within 30 days:** 1. The System displays an error message asking the patient to select another date range. 2. Resume at Main Flow Step 3.
6a	**No suitable appointment is found:** 1. The System displays an error message asking the patient to select another date range. 2. Resume at Main Flow Step 3.
8a	**The patient clicks on the "Cancel" button:** 1. No action is taken and the original appointment remains unchanged. 2. Use case ends.

USE CASE UC30: Internal calendar integration	
Goal in Context	To provide a centralized view where users can view all scheduled medical appointments within the application.
Scope & Level	Scope: Appointment Management. Level: Primary.
Preconditions	The user is authenticated and has an active account.
Success End Condition	The system successfully renders a calendar interface populated with the user's specific appointment data.
Failed End Condition	The calendar fails to load or appointments are not visible; the user receives an error message.
Primary, Secondary Actors	Primary: User. Secondary: Database.
Trigger	The user wants to consult their calendar.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR17	Internal calendar integration

DESCRIPTION

Step	Action
1	The user wants to consult their calendar.
2	The system displays the user's scheduled appointments in a calendar view, showing basic information.

EXTENSIONS

Step	Branching Action
2a	**No appointments found:** 1. The system doesn't find appointments to populate the calendar page. 2. The system displays an error message. 3. Use case ends
2b	**User changes view:** 1. The User selects a different time scale (Daily, Weekly, or Monthly). 2. The System re-formats the displayed data according to the selected scale.

USE CASE UC31: Appointment export	
Goal in Context	To provide users with a dynamic synchronization link that displays their medical appointments within external calendar applications (Google, Apple, Outlook).
Scope & Level	Scope: External Integration. Level: Primary.
Preconditions	The user is authenticated and has at least one scheduled or historical appointment record.
Success End Condition	A unique, secure URL is generated and provided to the user for external calendar synchronization.
Failed End Condition	Calendar export fails; the user is notified of the error.
Primary, Secondary Actors	Primary: User. Secondary: External Calendar Provider (e.g., Google Calendar Service).
Trigger	The user wants to export their calendar with an external application.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR18	Appointment Export

DESCRIPTION

Step	Action
1	The user wants to export their calendar with an external application.
2	The System generates a unique, non-guessable URL containing a secure access token.
3	The System presents the URL to the user.
4	The User uses the URL to import the calendar into their external application.

EXTENSIONS

Step	Branching Action
4a	**Token Expired/Invalid:** 1. The System receives a request from an external application with an invalid or expired token. 2. The System returns a synchronization error. 3. Use case ends.

USE CASE UC32: Payment management	
Goal in Context	To provide a secure, real-time financial transaction to pay upfront for medical services.
Scope & Level	Scope: Billing and Payment management. Level: Primary.
Preconditions	The patient is logged in and has booked a medical appointment.
Success End Condition	A payment authorization is received, a transaction ID is logged, and the appointment is marked as paid
Failed End Condition	The payment fails and the system remains unchanged
Primary, Secondary Actors	Primary: Patient. Secondary: Payment Gateway (e.g., Stripe, PayPal, Adyen), Database.
Trigger	The user wants to pay a booked appointment.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR19	Payment management

DESCRIPTION

Step	Action
1	The System retrieves the base price for the selected medical service.
2	**Include: Exemption application during payment.** The System calculates and applies health exemptions to the price.
3	**Include: Insurance application.** The System calculates and applies any valid insurance coverage to the price.
4	The user is presented with a payment summary showing the final amount.
5	The User selects a payment method (Credit Card, Google Pay, or Apple Pay).
6	The User enters credentials for the selected payment method.
7	The System logs the transaction ID, along with applied insurance or exemption deductions, and updates the appointment status.
8	The System displays a payment confirmation message to the user.

EXTENSIONS

Step	Branching Action
7a	**Transaction Declined:** 1. The Gateway returns a declined error. 2. The User is prompted with the error message. 3. Use case ends.
7b	**Session Timeout:** 1. The User abandons the payment page or the gateway connection drops. 2. The System notifies the user of the session failure. 3. Use case ends.

USE CASE UC33: Exemption application during payment	
Goal in Context	To automatically calculate and apply discounts based on exemptions that apply to the patient's profile.
Scope & Level	Scope: Billing and Payment management. Level: Subfunction.
Preconditions	The patient is logged in, has initiated a payment flow, and the initial price is known.
Success End Condition	The total cost is correctly reduced by the applicable percentage based on the patient's exemptions.
Failed End Condition	The system defaults to the full price; no discounts are applied.
Primary, Secondary Actors	Primary: System. Secondary: Patient, Database.
Trigger	The use case is triggered automatically when a patient wants to pay an examination. (**UC32**)

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR20	Exemption application during payment

DESCRIPTION

Step	Action
1	The System identifies the specific treatment associated with the requested appointment.
2	The System verifies the validity and applicability of each found exemption against the selected treatment for the specific patient.
3	The System calculates the cumulative discount and applies it to the base price.
4	The System returns the final discounted price and specific discount details to the calling use case.

EXTENSIONS

Step	Branching Action
2a	**No Exemptions registered:** 1. No applicable exemption is found for the patient or the specific treatment. 2. The System applies the full price (0% discount). 3. Resume at Main Flow Step 4.
2b	**Expired Exemption:** 1. The System finds an exemption that has passed its validity date. 2. The discount from that specific exemption is not applied. 3. Resume at Main Flow Step 4.
3a	**Price goes to 0 or below:** 1. After applying the exemptions, the resulting price is calculated as ≤ 0. 2. The System sets the final discounted price to exactly 0. 3. Resume at Main Flow Step 4.

USE CASE UC34: Insurance application during payment	
Goal in Context	To automatically calculate and apply discounts based on insurance policies that apply to the patient's profile.
Scope & Level	Scope: Billing and Payment management. Level: Subfunction.
Preconditions	The patient is logged in, has initiated a payment flow, and the initial price is known.
Success End Condition	The total cost is correctly reduced by the applicable percentage based on the insurance coverage.
Failed End Condition	The system defaults to the full price; no insurance discount is applied.
Primary, Secondary Actors	Primary: System. Secondary: Patient, Database.
Trigger	The use case is triggered automatically when a patient wants to pay an examination. (**UC32**).

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR21	Insurance application during payment

DESCRIPTION

Step	Action
1	The System identifies the specific treatment associated with the requested appointment.
2	The System verifies the validity and applicability of each active insurance policy against the selected treatment for the patient.
3	The System calculates the cumulative discount and applies it to the base price.
4	The System returns the final discounted price and specific coverage details to the calling use case.

EXTENSIONS

Step	Branching Action
2a	**No Insurance policy registered:** 1. No suitable or active insurance policy is found for the patient. 2. The System applies the full price. 3. Resume at Main Flow Step 4.
2b	**Expired Insurance policy:** 1. The System finds a policy that has passed its validity date. 2. The discount from that policy is not applied to the transaction. 3. Resume at Main Flow Step 4.
3a	**Price goes to 0 or below:** 1. After applying the insurance discount, the resulting price is calculated as ≤ 0. 2. The System sets the final price to exactly 0. 3. Resume at Main Flow Step 4.

USE CASE UC35: Automatic appointment booking from prescription	
Goal in Context	To automatically initiate the booking process from the prescriptions page.
Scope & Level	Scope: Appointment Management. Level: Primary.
Preconditions	The patient is authenticated and has at least one active digital prescription stored in the system.
Success End Condition	The system extracts the relevant prescription data initiates the booking process with pre-filled information about the treatment.
Failed End Condition	The initiation fails.
Primary, Secondary Actors	Primary: Patient. Secondary: Database.
Trigger	Patient wants to directly book an appointment for a specific prescription.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR22	Automatic appointment booking from prescription
NFR1	Algorithm performance

DESCRIPTION

Step	Action
1	Patient wants to directly book an appointment for a specific prescription..
2	The System checks whether the prescription has already been used and verifies its validity period to ensure it has not expired.
3	The System reads the treatment category from the prescription data.
4	The System initiates the appointment booking process with the treatment field pre-filled.
5	The System tags the prescription as fulfilled, blocking other appointments related to the same prescription.

EXTENSIONS

Step	Branching Action
2a	**Prescription Already Used:** 1. The System detects that the prescription has already been fulfilled by a previous booking. 2. The Patient is prompted with an error message. 3. Use case ends.
2b	**Expired Prescription:** 1. The System identifies that the prescription is expired. 2. The Patient is prompted with an error message. 3. Use case ends.
4a	**Redirect Failure:** 1. The System fails to redirect the Patient to the book appointment tab. 2. The Patient is prompted with an error message. 3. Use case ends.

USE CASE UC36: Patient insurance registration via unique code	
Goal in Context	To enable users to securely link their private insurance coverage to their account profile using a provider-issued identification code.
Scope & Level	Scope: Patient Profile Management. Level: Primary.
Preconditions	The patient is logged-in. The patient has a valid insurance policy code.
Success End Condition	The policy is successfully linked to the user's profile for future use in billing, typically through **Insurance application during payment**.
Failed End Condition	The unique code is unrecognized, expired, or already associated with a different primary account.
Primary, Secondary Actors	Primary: Patient. Secondary: Database.
Trigger	The Patient wants to link an existing insurance policy to their hospital profile using their policy code.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR23	Patient insurance registration via unique code

DESCRIPTION

Step	Action
1	The Patient wants to link an existing insurance policy to their hospital profile using their policy code.
2	The Patient enters their unique insurance code.
3	The System validates the insurance code.
4	The System displays a summary of the retrieved policy for the Patient to confirm.
5	The Patient confirms that the displayed insurance information is correct.
6	The System links the insurance policy to the Patient's profile.

EXTENSIONS

Step	Branching Action
2a	**Invalid Code:** 1. The System finds no match for the provided code. 2. The System displays an error message. 3. Use case ends.
2b	**Duplicate Code:** 1. The System detects that the code is already linked to another active User ID 2. The System displays an error message. 3. Use case ends.
2c	**Expired Policy Record:** 1. The System finds the code, but the database indicates that the policy coverage ended in the past. 2. The System prevents the insurance registration. 3. The Patient is notified to update their policy. 4. Use case ends.
4a	**User Cancels:** 1. The Patient sees the summary and realizes it refers to the wrong insurance plan. 2. The Patient wants to cancel. 3. The System discards the lookup results and returns the Patient to the dashboard. 4. Use case ends.

USE CASE UC37: AI Assistant	
Goal in Context	To provide users with an intelligent, conversational interface that simplifies navigation and assists in completing complex tasks such as booking or insurance registration.
Scope & Level	Scope: User Support. Level: Primary.
Preconditions	The patient is authenticated.
Success End Condition	The user receives accurate, context-specific answers or successfully completes a task following the AI Assistant's guidance.
Failed End Condition	The AI Assistant fails to interpret the user's intent or provides information that contradicts system rules.
Primary, Secondary Actors	Primary: Patient. Secondary: AI Large Language Model API, Database.
Trigger	The user needs to perform an action and doesn't know how or where to do it.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR24	AI assistant

DESCRIPTION

Step	Action
1	The user needs to perform an action and doesn't know how or where to do it.
2	The System initializes the AI Chat Interface and greets the Patient, potentially offering a proactive tip based on the current page.
3	The Patient inputs a natural language query.
4	The System retrieves the Patient's user context.
5	The AI Assistant processes the query by cross-referencing the internal knowledge base with the Patient's context.
6	The AI Assistant explains the situation and provides a step-by-step resolution.
7	The AI Assistant presents a redirection for the first action.
8	The Patient selects the redirection.

EXTENSIONS

Step	Branching Action
4a	**Ambiguous Query:** 1. The AI Assistant cannot confidently identify the Patient's intent. 2. The System provides a list of "Did you mean...?" options. 3. Resume at Main Flow Step 2.
5a	**Policy Limitation or Privacy Restriction:** 1. The Patient asks for information that the AI Assistant is not authorized to share. 2. The AI Assistant politely declines the request. 3. Resume at Main Flow Step 2.

USE CASE UC38: Prescription consultation	
Goal in Context	To allow users to view a list of prescriptions issued to them by healthcare providers.
Scope & Level	Scope: Patient Portal. Level: Primary.
Preconditions	The user is logged into their account.
Success End Condition	The user successfully views the list of prescriptions and can identify their status and provider.
Failed End Condition	The system fails to retrieve prescription data; an error message is displayed.
Primary, Secondary Actors	Primary: Patient. Secondary: Database
Trigger	The User wants to see their prescriptions.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR25	Prescription consultation
NFR0	Security and Privacy
NFR5	Accessibility

DESCRIPTION

Step	Action
1	The User wants to see their prescriptions.
2	The System retrieves all prescriptions associated with the User's ID from the database.
3	The System displays a list of prescriptions showing high-level information (Date, Provider, Medication/Treatment name).

EXTENSIONS

Step	Branching Action
2a	**No Records Available:** 1. The System identifies that the user has no issued prescriptions. 2. The System displays an error message. 3. Use case ends.

USE CASE UC39: Detailed prescription access	
Goal in Context	To allow a patient to view the comprehensive details of a specific medical prescription.
Scope & Level	Scope: Patient prescription management. Level: Primary.
Preconditions	The user is logged in as a Patient. At least one prescription is associated with the patient's profile.
Success End Condition	The patient successfully views the full details of the selected prescription.
Failed End Condition	The prescription details are not displayed, and the patient remains on the prescription list.
Primary, Secondary Actors	Primary: Patient. Secondary: Healthcare Database.
Trigger	The patient wants to see details for a specific prescription assigned to them.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR26	Detailed Prescription access
NFR0	Security and privacy

DESCRIPTION

Step	Action
1	The patient wants to see details for a specific prescription assigned to them
2	The System displays a list of the Patient's prescriptions.
3	The Patient selects a specific prescription from the list.
4	The System retrieves the prescription data from the database.
5	The System displays the treatment name, prescribing Doctor, issue date, expiration date, priority level, and any associated notes.

EXTENSIONS

Step	Branching Action
3a	**Data retrieval error:** 1. The system fails to fetch details from the database. 2. An error message is displayed to the patient. 3. The patient is redirected back to the prescription list.

USE CASE UC40: Patient medical record access (Patient)	
Goal in Context	To enable a patient to access their medical records and organize them using various filters and sorting criteria.
Scope & Level	Scope: Patient medical history. Level: Primary.
Preconditions	The user is logged in as a Patient. The system has verified that the user is only accessing records linked to their specific ID.
Success End Condition	The patient views their filtered/sorted medical records and exam results.
Failed End Condition	The patient cannot access the records or sees an empty list due to a system error.
Primary, Secondary Actors	Primary: Patient. Secondary: Database.
Trigger	The patient wants to access their medical records.

DESCRIPTION

Step	Action
1	The patient wants to access their medical records.
2	The system retrieves and displays all medical records and exam results for the authenticated patient.
3	Patient chooses to sort the records.
4	Patient selects a sorting criterion (type, date/time, doctor, or location).
5	The system reorders the records based on the selected parameter.
6	Patient selects an individual record or result to consult.
7	The system displays the full content of the selected record.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR27	Patient medical records and result access
NFR0	Security and privacy
NFR1	Algorithm performance

EXTENSIONS

Step	Branching Action
1a	**Access Violation Attempt:** 1. The system detects an attempt to access records not belonging to the logged-in ID. 2. Access is denied and a security warning is logged. 3. The use cases ends.
3a	**No records found:** 1. The system finds no records matching the patient's ID. 2. The system displays an error message.

USE CASE UC41: Account Deletion	
Goal in Context	To allow users to permanently delete their HealthMaxxin account, ensuring the complete removal of their personal data from the platform.
Scope & Level	Scope: Users account management. Level: Primary.
Preconditions	The user account exists and the user is logged in.
Success End Condition	The user account and all associated authentication and personal data are deleted.
Failed End Condition	The deletion request is aborted and the account remains unchanged.
Primary, Secondary Actors	Primary: User. Secondary: Database, Mail Service.
Trigger	The user want to delete their account.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR28	Accout Deletion
NFR0	Security and privacy

DESCRIPTION

Step	Action
1	The user deletes their account.
2	The System prompts the user with a confirmation message asking whether they really want to delete their account.
3	The user confirms.
4	The System deletes the account and personal data, keeping a local copy for 30 days.
5	The System automatically sends a confirmation email to the user's registered email address after the successful deletion of the account.

EXTENSIONS

Step	Branching Action
2a	**Deletion Aborted:** 1. The user cancels during the confirmation step. 2. The System aborts the deletion request. 3. No changes are made to the account or personal data. 4. Use case ends.

USE CASE UC42: Medical record access (Doctor)	
Goal in Context	To allow a doctor to retrieve a patient's clinical history by either manually entering their Fiscal Code or scanning their Sanitary Card barcode.
Scope & Level	Scope: Accessing Patient Clinical History. Level: Primary Task.
Preconditions	The patient is registered in the system.
Success End Condition	The Doctor successfully retrieves and views the patient's relevant clinical history.
Failed End Condition	No records are found, the input is invalid, or access is restricted; the System shows an error and no data is retrieved.
Primary, Secondary Actors	Primary: Doctor. Secondary: Database.
Trigger	The Doctor wants to search for clinical records.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR31	Patient medical record access via manual ID entry
FR32	Patient medical record access via barcode scan
NFR0	Security and privacy
NFR1	Algorithm performance

DESCRIPTION

Step	Action
1	The Doctor wants to search for clinical records.
2	The Doctor inputs the Patient's Fiscal Code manually or the manual ID entry.
3	The System validates the format of the input.
4	The System searches the database for matching Patient records.
5	The System retrieves only the clinical information relevant to the booked visit (the Patient's full history is restricted due to privacy concerns).
6	The Doctor views the retrieved clinical information.

SUB-VARIATIONS

Step	Branching Action
1a	**Barcode Scan Input:** 1. The Doctor scans the barcode from the Patient's Sanitary Card instead of typing manually. 2. The System processes and decodes the barcode to extract the Fiscal Code. 3. Resume at Main Flow Step 2.

EXTENSIONS

Step	Branching Action
1a.1	**Barcode Not Readable:** 1. During a barcode scan attempt, the System fails to read or decode the barcode. 2. The System displays an error message. 3. The Doctor is prompted to enter the Patient information manually. 4. Resume at Main Flow Step 1.
2a	**Invalid Input Format:** 1. The System detects that the entered or scanned Fiscal Code has an invalid format. 2. The System displays an error message. 3. The Doctor corrects the input and submits it again. 4. Resume at Main Flow Step 1.
3a	**Patient Not Found:** 1. The System does not find any matching Patient records. 2. The System displays a message indicating that the Patient is not present in the system. 3. Use case ends.
4a	**No Relevant Record Found:** 1. The System detects that the Patient has no clinical records relevant to the current context or previous visits with the Doctor. 2. The System displays a message indicating that no clinical records are available. 3. Use case ends.
4b	**Access Restricted Due to Privacy:** 1. The System detects that access to the Patient's clinical history is restricted due to privacy settings. 2. The System displays an authorization message indicating that the Doctor cannot access the Patient's clinical information. 3. Use case ends.

USE CASE UC43: Doctor chronological report	
Goal in Context	To allow a doctor to view a chronological report of their performed examinations from the last year for review and clinical decision-making.
Scope & Level	Scope: Doctor Examination History Reporting. Level: Primary.
Preconditions	The doctor is authenticated in the system and at least one examination or visit record exists for the doctor in the last year.
Success End Condition	The Doctor successfully views the chronological report of examinations up to the last year.
Failed End Condition	The report cannot be generated due to a system error or data retrieval failure.
Primary, Secondary Actors	Primary: Doctor. Secondary: Database.
Trigger	The use case is triggered when the Doctor wants to view the chronological report from their dashboard.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR34	Doctor chronological report

DESCRIPTION

Step	Action
1	The Doctor wants to view the chronological report.
2	The System retrieves the Doctor's examinations from the past year.
3	The System shows a chronological list of the Doctor's performed visits from the past year.
4	The Doctor can filter the list for a more detailed view.

EXTENSIONS

Step	Branching Action
2a	**No Examination Records Found:** 1. The System finds no examination records for the Doctor from the past year. 2. The System displays an empty report or a message indicating that no records are available. 3. Use case ends.

USE CASE UC44: Assign doctor shift	
Goal in Context	To enable the Head of Department to specify shift patterns for doctors.
Scope & Level	Scope: Hospital structures handling. Level: Primary.
Preconditions	The user is logged in as a Head of Department. The doctors are registered in the system and assigned to the department.
Success End Condition	The shift patterns for doctors are successfully updated.
Failed End Condition	The system remains unchanged; shift patterns stay as they were.
Primary, Secondary Actors	Primary: Head of Department. Secondary: Database, Mail Service.
Trigger	The Head of Department wants to specify shift patterns for doctors.

RELATED FUNCTIONAL / NON-FUNCTIONAL REQUIREMENTS

FR ID	Title
FR8.2	Doctor shift assignement

DESCRIPTION

Step	Action
1	The Head of Department wants to specify shift patterns for doctors.
2	The Head of Department selects a specific doctor.
3	The Head of Department enters the new shift pattern for the selected doctor.
4	The Head of Department saves.
5	The System performs validity checks.
6	The shift pattern is updated and a notification is sent to the doctor via the Mail Service.

EXTENSIONS

Step	Branching Action
5a	**Shift pattern invalid:** 1. The System identifies the shift pattern as invalid (e.g., overlapping shifts). 2. The System returns an error. 3. Resume at Main Flow Step 3.

Activity Diagram

1. Book Appointment

Here there's the entire flow of the booking of an appointment.

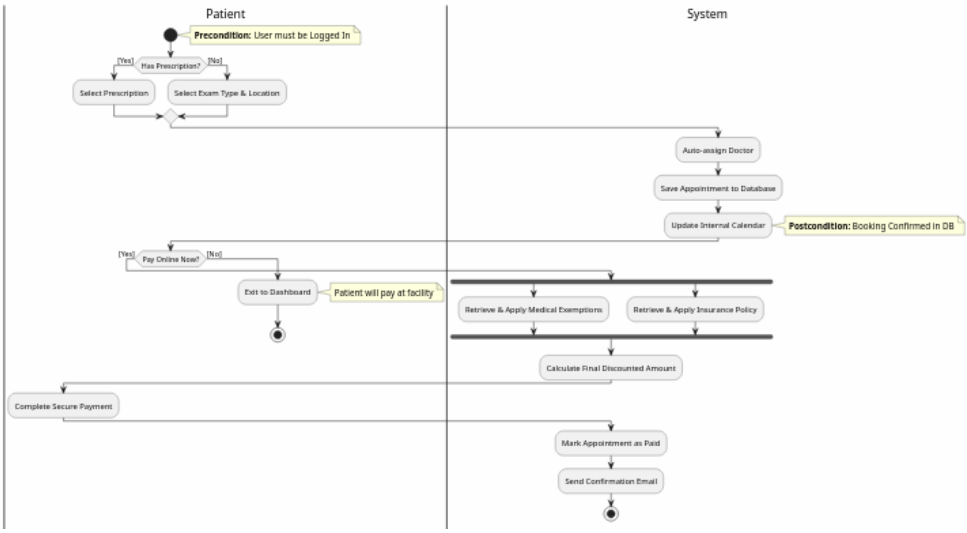

Figure 1: Activity Diagram for Book Appointment

- **Using:** Automated doctor assignment, Appointment booking, Payment management, Exemption application during payment, Insurance application during payment, Automatic appointment booking from prescription

2. Calendar Management

Here there's the entire flow of the management of a calendar.

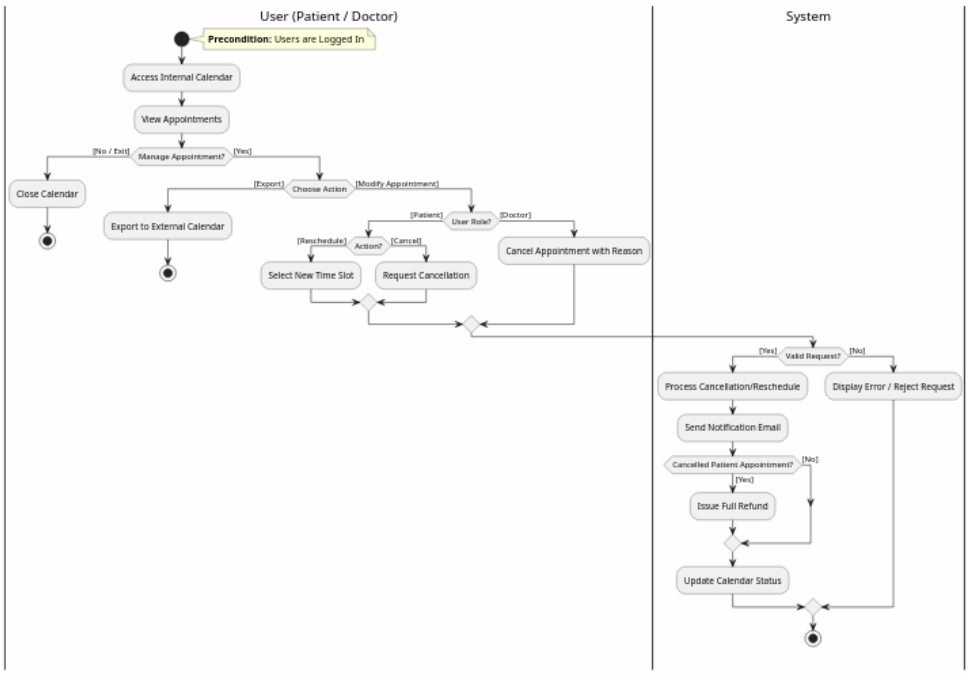

Figure 2: Activity Diagram for Calendar Management

- **Using:** Patient appointment view, Patient appointment cancellation, Patient appointment rescheduling, Internal calendar integration, HoD appointment view, Hospital staff appointment cancellation

3. Admin Actions

Here the flow of all the admin's possible actions.

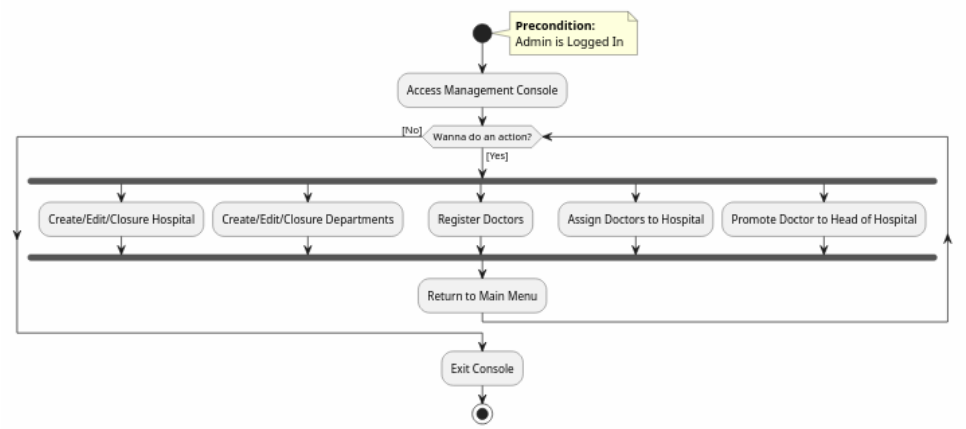

Figure 3: Activity Diagram for Admin Actions

- **Using:** Hospital and Department creation, Hospital and Department edit, Hospital and Department closure, Doctor registration, Doctor assignment to a hospital, Promote doctor to HoH

4. Head of Hospital Actions

Here there's the entire flow of all the Head of Hospital's possible actions.

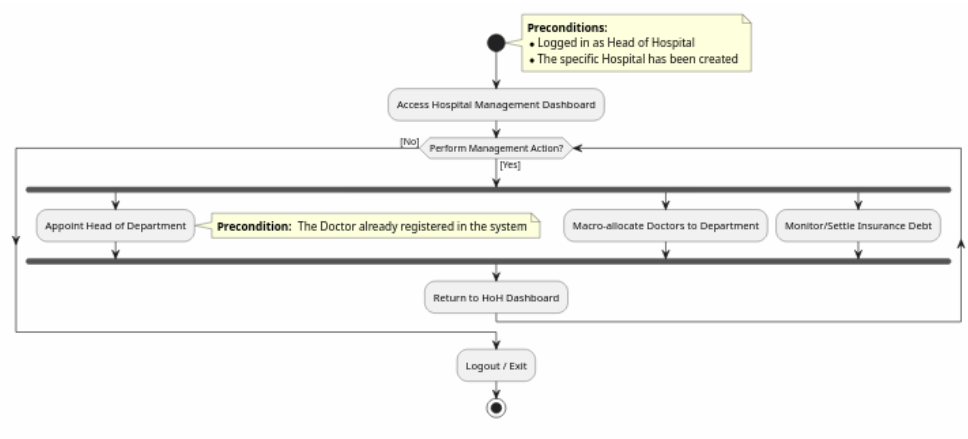

Figure 4: Activity Diagram for Head of Hospital Actions

- **Using:** Appoint HoD, Macro-allocate Doctors to Deparments, Insurance debt tracking and reporting, Insurance debt History, Manual settlement of insurance debt

5. Head of Department Actions

Here there's the entire flow of all the Head of Department's possible actions.

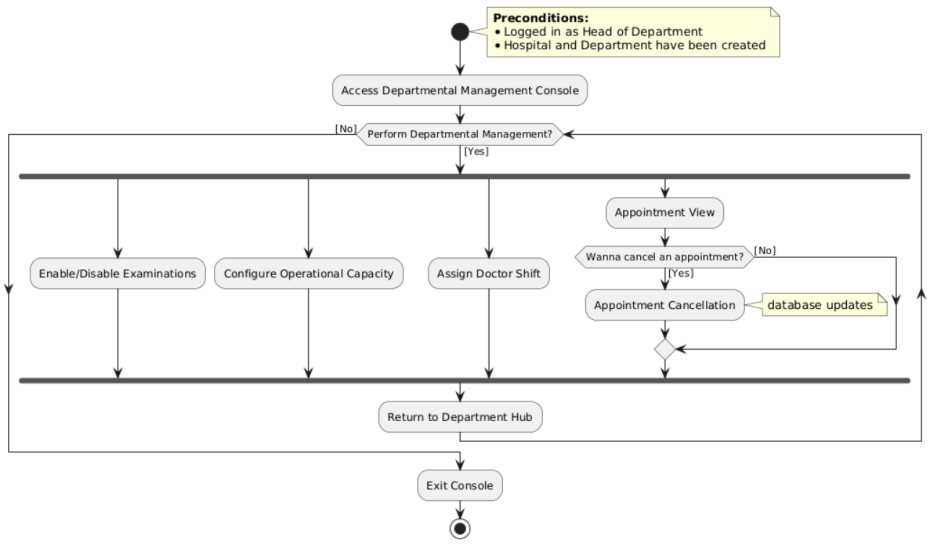

Figure 5: Activity Diagram for Head of Department Actions

- **Using:** Enable/Disable Examinations, Configure Operational Capacity, Assign Doctor Shift, Appointment View, Appointment Cancellation

6. Doctor Operations

Here's the flow of the doctor's action to manage the their patients' account (exemptions/examinations/prescriptions).

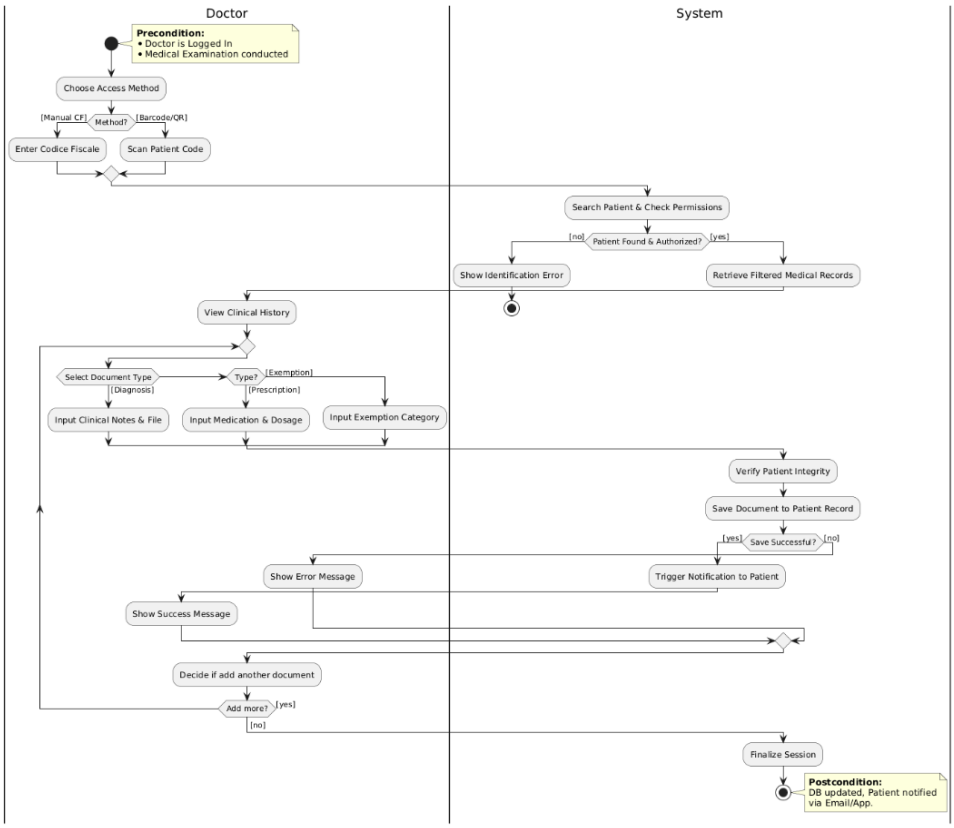

Figure 6: Activity Diagram for Doctor Operations

- **Using:** Access Medical Record via CF or Scan, Insert Prescription, Insert Diagnosis, Insert Exemption

7. Account Lifecycle

Here's the activity flow of all the action related to the creation/management/deletion of an user account (which can be doctors or patients).

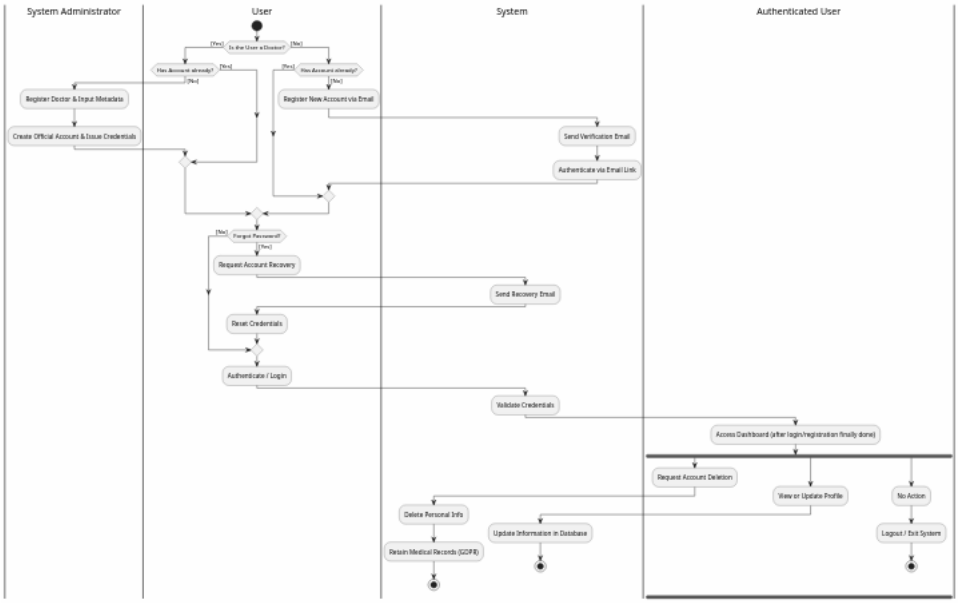

Figure 7: Activity Diagram for Account Lifecycle

- **Using:** Doctor Registration, User Registration and Authentication via Email, Password Recover, Login, Delete Account, Update Account

Sequence Diagram

1. Reschedule Appointment

Here there's the sequence diagram of the reschedule of an appointment.

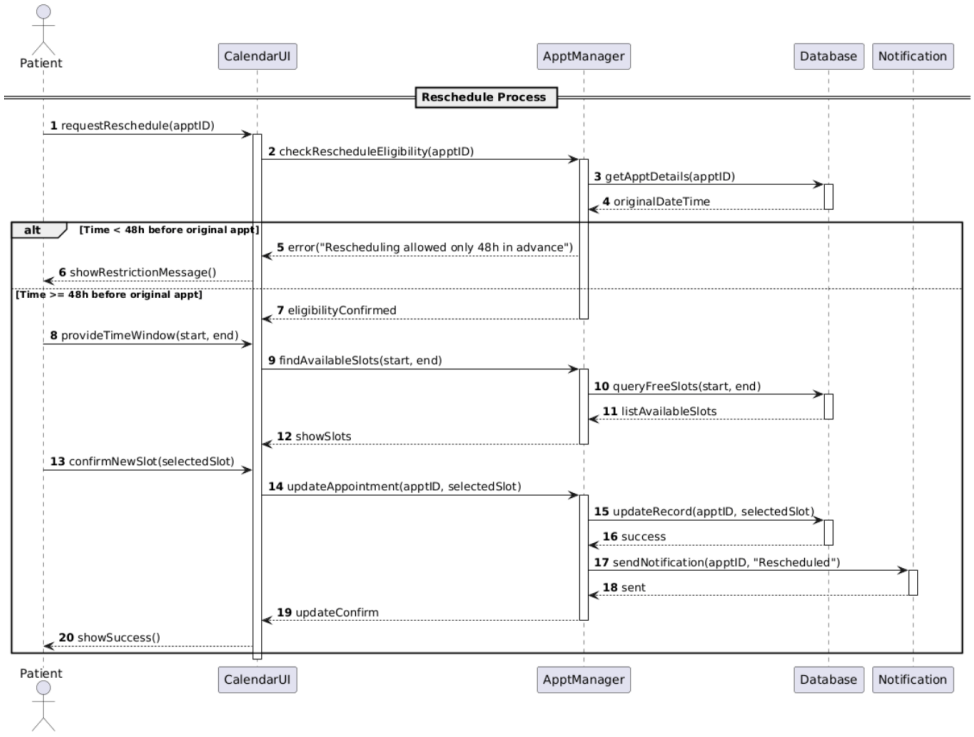

Figure 8: Sequence Diagram for Reschedule Appointment

- **Using:** Patient Appointment Rescheduling

2. Cancel Appointment

Here is the sequence diagram of the cancel of an appointment, from both patient or hospital staff point of view.

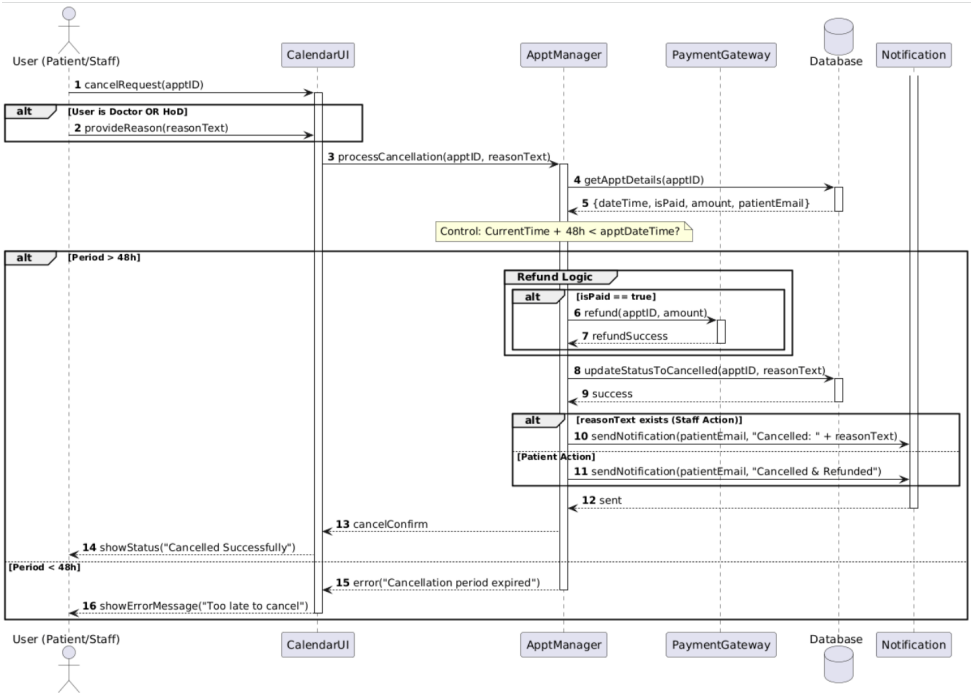

Figure 9: Sequence Diagram for Cancel Appointment

- **Using:** Patient cancel appointment, Hospital staff cancel appointment

3. Doctor Upload Document

Here is the sequence activity related to the upload of documents by doctor into the patient's medical record.

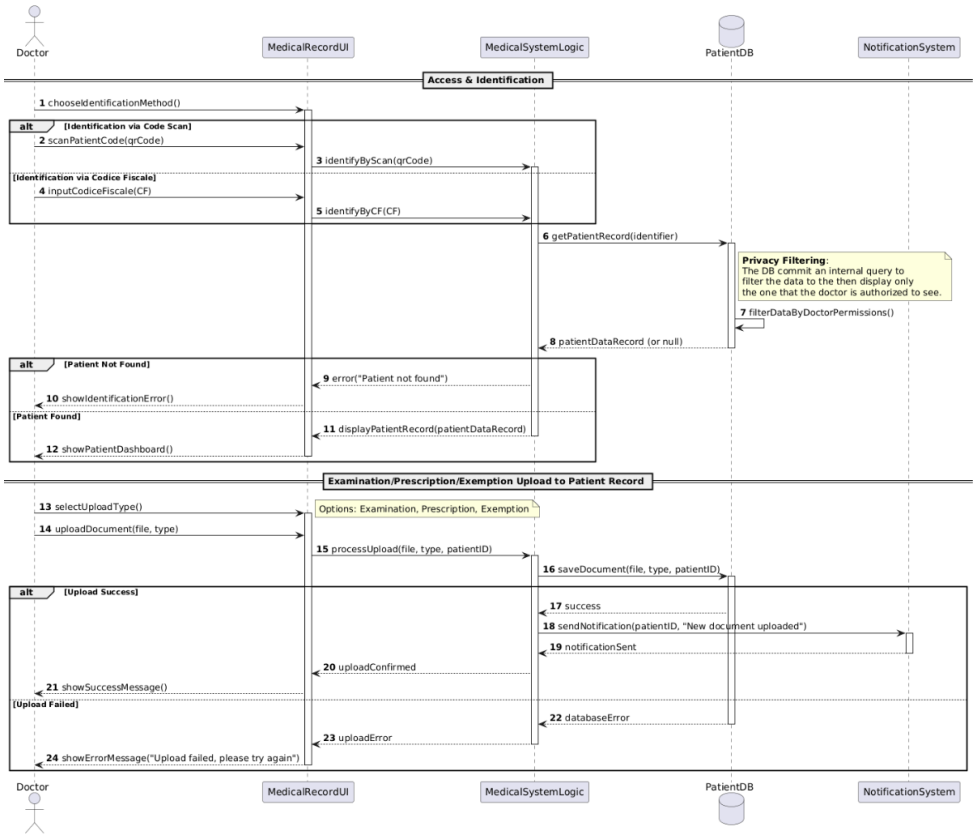

Figure 10: Sequence Diagram for Doctor Upload Document

- **Using:** Access Medical Record via CF or Scan, Insert Prescription, Insert Diagnosis, Insert Exemption

4. Registration User

Here is the sequence diagram about the registration of an user (that can be either a doctor or a patient).

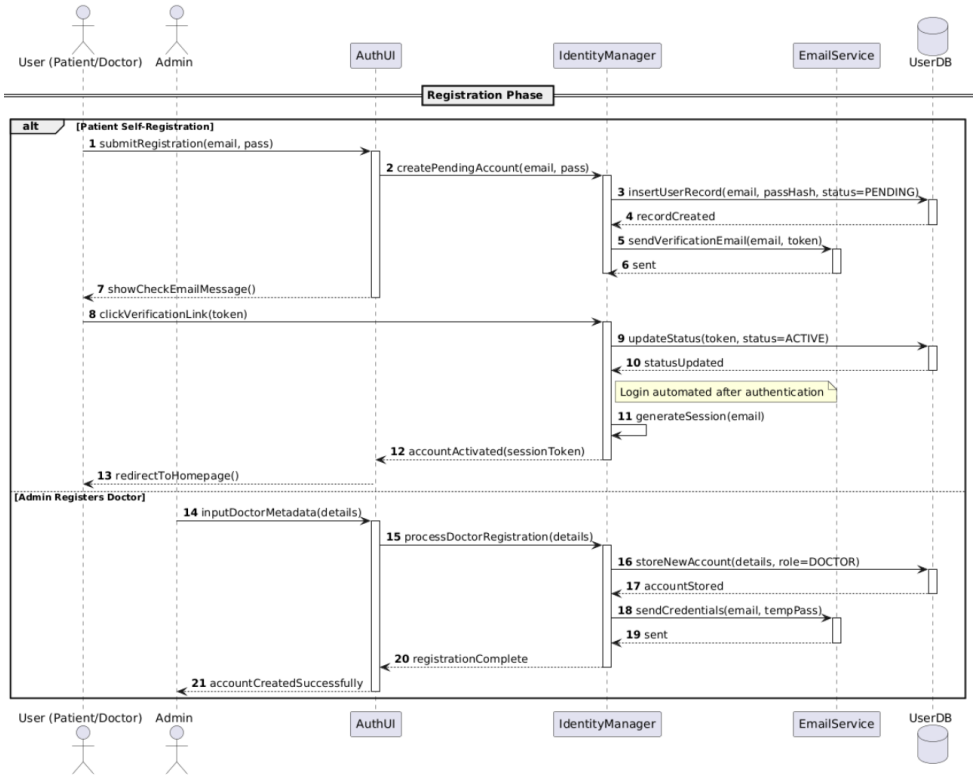

Figure 11: Sequence Diagram for Registration User

- **Using:** Doctor Registration, User Registration and Authentication via Email

5. Book Appointment

Here the sequence diagram about the book of an appointment by a patient.

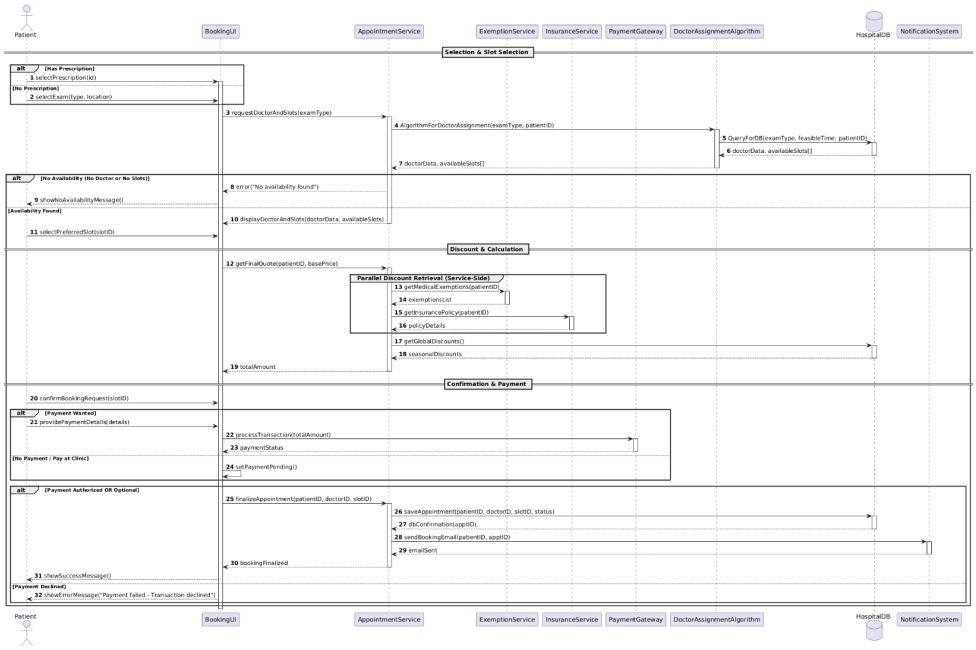

Figure 12: Sequence Diagram for Book Appointment

- **Using:** Automated doctor assignment, Appointment booking, Payment management, Exemption application during payment, Insurance application during payment, Automatic appointment booking from prescription

6. Account Login and Password Recovery

Here is the sequence diagram about the password recovery, that need the context of the user login, so we decided to implement them in a unique sequence diagram.

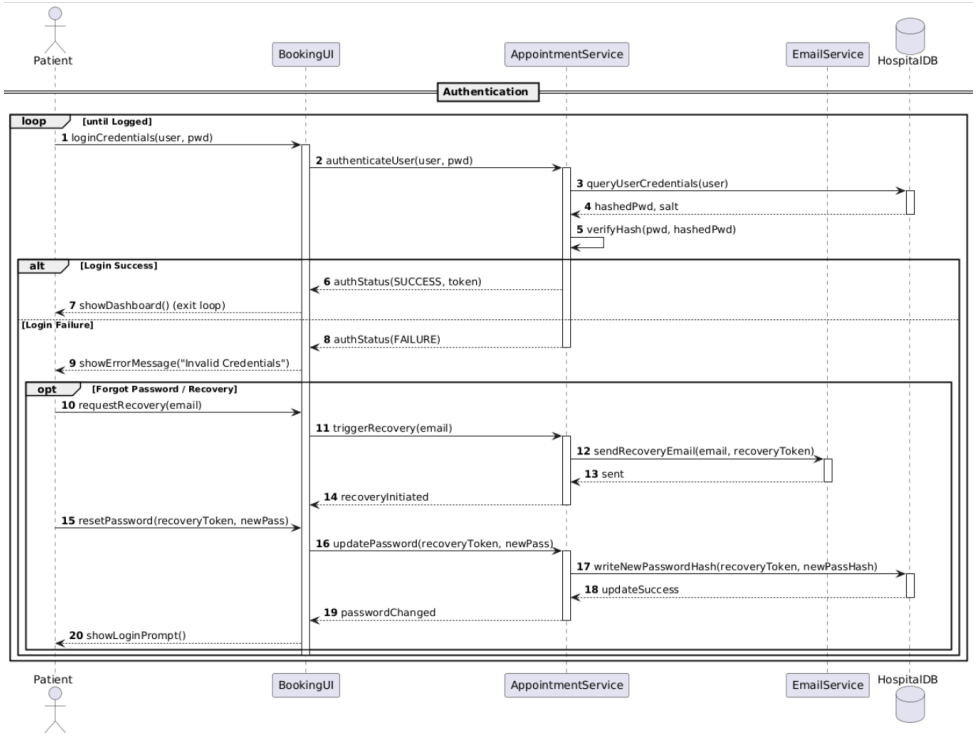

Figure 13: Sequence Diagram for Login and Password Recovery

- **Using:** Password Recover, Login

7. Account Management

Here the sequence diagram about the operations related to the edit and deletion of an user account (that can be either a doctor or a patient).

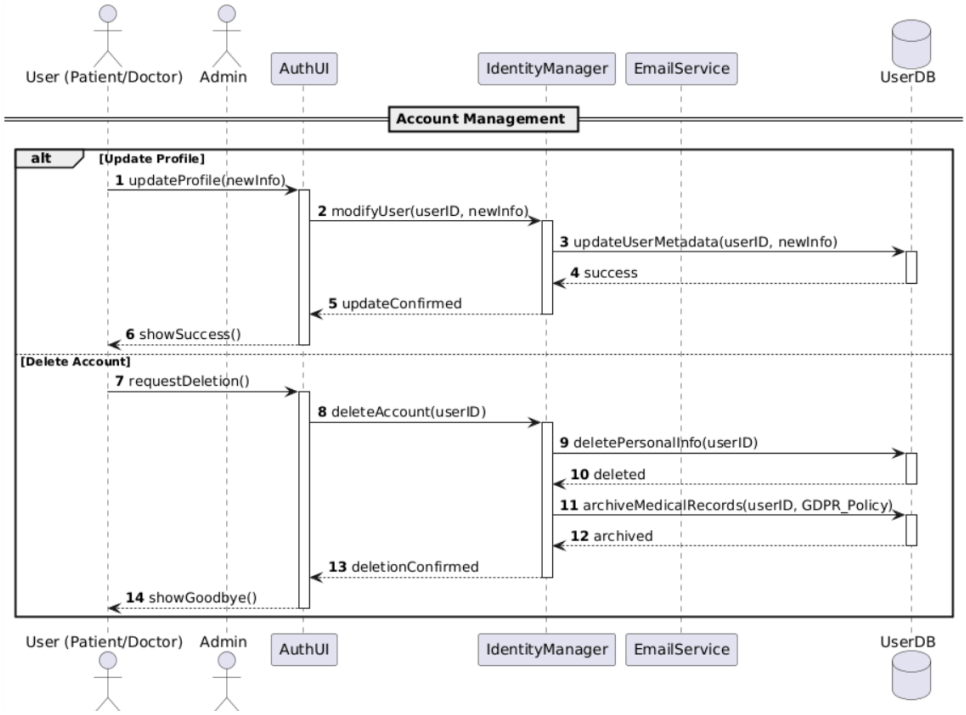

Figure 14: Sequence Diagram for Account Management Operations

- **Using:** Delete Account, Update Account

Class Diagram

For readability purposes, the class diagram has been attached separately as a PDF document.